Geometric State Manifolds for Transferable, Data-Efficient Dynamic-System Learning

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Abstract

Flat timestamped telemetry, with rows as time and columns as sensor channels, is the dominant input representation for models that monitor, predict, or control physical systems, yet the coupled states of any such machine do not fill the observation box. They live on a constrained, lower-dimensional manifold whose geometry is exactly the structure that makes a system learnable and its failures detectable. We introduce the Index of Operators and Operations, a representation that turns raw sensor streams into a hierarchical geometric graph, where each hardware primitive is a Riemannian manifold with a physical operating envelope, the primitives are glued at their mechanical interfaces, and message-passing networks read the result. On NASA C-MAPSS turbofan telemetry across one to six operating conditions and multiple fault modes, the graph network dominates flat MLP and GRU baselines in the low-data regime on every suite, a regime-conditional extension preserves the advantage under heavy operating-condition transients, and the same graph transfers zero-shot across a motor-family swap where the flat baseline degrades 2.3 times more. A conditional residual on the discovered graph detects incipient faults a median of 89.7 seconds before functional failure where marginal-correlation detectors fire only at failure, and real bearing vibration confirms the method ports outside simulation. The transferable lesson for physical-science machine learning is that inter-channel coupling is a prior worth encoding in the architecture, because it buys data efficiency and failure detection that marginal statistics cannot supply, and on an instrumented machine it can be discovered from telemetry rather than declared by hand.

1 Introduction

Industrial and robotic systems are instrumented with dozens to hundreds of correlated sensor channels, and the learned models that predict remaining useful life, schedule maintenance, or close a control loop are usually trained on flattened windows of that telemetry. A flat regressor therefore has to re-derive from raw statistics the physical coupling that generated the data, and the cost of discarding that structure shows up in three ways. It is data inefficient, because a system with d sensors and k hidden physical constraints lives on a manifold of intrinsic dimension at most $d - k$ while a flat regressor explores all of $\mathbb{R}^d$. It is brittle, because when hardware changes the marginal statistics of every correlated channel shift together. And it is blind to the degradation modes that matter most, because bearing wear, coupling loss, and gain drift change the strength and dynamics of inter-channel coupling rather than their marginal levels.

Those three costs share one root cause, which is that the physical wiring of a machine, meaning which variables are coupled and how strongly, is the prior its learned models should consume and flat telemetry throws it away. This paper formalizes that coupling structure as a geometric graph and shows that networks reading its topology inherit transferability, sample efficiency, and early warning that flat data injection does not provide, through the same mechanism that makes weight sharing useful elsewhere, since the architecture supplies a prior and the prior here is the physics of coupling.

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Two measured results carry the paper. The first is that the graph is a sample-efficiency prior. On NASA C-MAPSS turbofan telemetry the graph network beats flat baselines at 2% of the training data on all three suites, including the multi-condition suites where a single model has to cover six operating regimes, and the advantage disappears at saturation, which is the expected signature of a structural prior rather than a capacity claim. The second is that the same object carries failures that marginal statistics cannot see. On runs where a bearing fault ramps in with the wear channel hidden, a conditional residual on the discovered graph detects every failure a median of 89.7 seconds before functional failure, while the standard correlation-drift detector catches 74% of them and fires essentially at failure.

2 Problem formulation

Operators as manifolds with an operating envelope. An *operator class* c is a triple $(\mathcal{M}_c,\, g_c,\, \mathcal{E}_c)$, where $\mathcal{M}_c \subseteq \mathbb{R}^{d_c}$ is a smooth state manifold for the hardware primitive, g_c is a Riemannian metric under which geodesic distance measures operational rather than merely numeric distance, and $\mathcal{E}_c \subset \mathcal{M}_c$ is the compact operating envelope of admissible states. For a motor the natural state is (ω, I, T_w), angular velocity, armature current, and winding temperature, with envelope $\mathcal{E}_c = \{(\omega, I, T_w) : |I| \le I_{\max}(\omega),\, T_w \le T_{\max}\}$. The envelope is what makes the manifold physical rather than geometric, since states outside it are not merely unlikely but inadmissible, and the geometry that matters for learning is its restriction to $\mathcal{E}_c$.

Composition at interfaces. A physical instance embeds $\mathcal{M}_c$ into the machine's global observation space $\mathbb{R}^d$ and exposes an interface signature $\partial c \subseteq \{1, \dots, d\}$, the channels through which it couples to its neighbors. Given instances $\{(\mathcal{M}_{c_i}, \phi_i)\}_{i=1}^{n}$, the Index of Operations is the fibered composition

$$\mathcal{I}_{\mathrm{ops}} = \bigsqcup_{i=1}^{n} \phi_i(\mathcal{M}_{c_i}) / \sim, \tag{1}$$

where $\sim$ identifies coupled channels at each physical interface, so motor shaft speed is identified with actuator lead-screw pitch, an identification that is a physical fact rather than a learned one. Two properties make (1) the right object for a systems setting. It is a graph, because nodes are operator instances or channels and edges are physical interfaces, and that graph is the topological skeleton a network reads. It is also modular, because replacing an instance with another class that shares the same interface signature edits only the local block and leaves the global topology unchanged, so a shared-weight message-passing network over $\mathcal{I}_{\mathrm{ops}}$ is a valid network over the modified $\mathcal{I}'_{\mathrm{ops}}$ with *the same* parameters.

Discovering the graph from telemetry. For a machine observed only through telemetry, as an instrumented turbofan is, the coupling graph has to be discovered rather than declared. With T z-scored samples $z(t) \in \mathbb{R}^d$ we retain edge (i, j) when its partial correlation exceeds a threshold τ, where $\Theta = \Sigma^{-1}$ is a ridge-regularized precision matrix and

$$\pi_{ij} = -\frac{\Theta_{ij}}{\sqrt{\Theta_{ii}\Theta_{jj}}}, \qquad \Sigma = \frac{1}{T-1}\sum_{t} \big(z(t) - \bar{z}\big)\big(z(t) - \bar{z}\big)^{\top} + \lambda I. \tag{2}$$

Partial correlation encodes conditional independence exactly only under joint Gaussianity, so we also build a non-parametric variant that estimates $\mathrm{CMI}(z_i; z_j \mid z_{\setminus\{i,j\}})$ through k-nearest-neighbor residualization and a Kraskov-Stögbauer-Grassberger estimator [Kraskov et al., 2004], which yields a conservative but usable graph without the linear assumption. The formal object fixes what a discovered graph has to preserve when an instance is swapped, which is the interface pattern rather than the channel identity.

Message passing on the discovered graph. Node features are per-window channel summaries, the mean, slope, minimum, and maximum, so $F{=}4$. With adjacency $(\mathcal{E}, w)$ from (2) or the non-parametric estimator, a shared-weight message-passing network aggregates neighbor states weighted by conditional coupling,

$$h_i^{(0)} = \mathrm{enc}(x_i), \qquad m_i^{(l)} = \sum_{j \in \mathcal{N}(i)} \frac{w_{ij}}{\deg(i)}\, \widetilde{h}_j^{(l)}, \qquad h_i^{(l+1)} = \mathrm{GELU}\Big(W_2^{(l)}\big[h_i^{(l)} \| m_i^{(l)}\big]\Big). \tag{3}$$

Because every weight is shared across nodes, the network is a function of the graph rather than of node identities, so the same trained parameters apply to any machine whose channels match the

Table 1: Graph network against baselines on NASA C-MAPSS held-out engines, in cycles, at $\tau = 0.5$, with n training windows in parentheses. Low is 2% of the training data and full is 100%. The no-edge column is the same network with the adjacency removed, so it isolates topology from architecture.

Suite	data	IOO-GNN	Reg-GNN	no-edge	MLP	GRU
FD001	2% (138)	36.5	41.8	41.7	51.0	76.2
(1 cond.)	100% (6928)	21.7	22.1	22.0	19.4	19.4
FD002	2% (371)	44.0	41.8	64.5	62.2	79.4
(6 cond.)	100% (18587)	41.8	41.3	41.8	23.1	22.0
FD004	2% (433)	43.8	41.8	70.5	68.0	86.1
(6 cond.)	100% (21696)	41.1	39.0	41.1	22.7	20.4

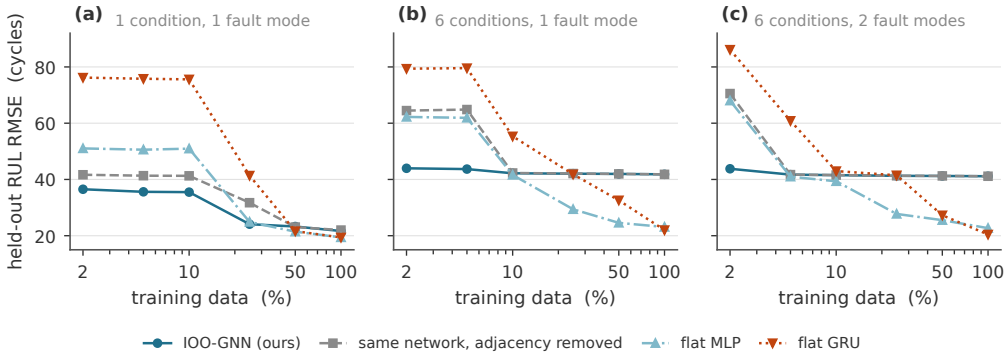


Figure 1: Sample efficiency of the discovered graph over the same $n = 6$ training fractions as Table 1, values repeated there: panel (a) is FD001 with one operating condition, panel (b) is FD002 and panel (c) is FD004, both with six operating conditions. The graph network leads on every suite where data is scarce and the MLP and GRU baselines win at saturation, which is the signature of a structural prior rather than of capacity.

feature schema and whose adjacency is given. Per-node tasks return $h_i^{(L)}$ through a shared head and graph-level tasks such as remaining-useful-life regression mean-pool over nodes. A regime-conditional extension gates the edge weights on the per-window operating-condition vector u, so message passing itself adapts to the current dynamical regime instead of averaging over all of them.

3 Experiments

Result 1: the graph is a sample-efficiency prior, and it fails at saturation exactly as a prior should. Table 1 reports held-out-engine remaining-useful-life RMSE in cycles on NASA C-MAPSS as the training fraction grows from 2% to 100%, across FD001 with one operating condition and one fault mode and FD002 and FD004 with six operating conditions, FD004 also with two fault modes [Saxena et al., 2008a]. The graphs are discovered from the training telemetry at $\tau = 0.5$, giving 105 edges over 15 sensors for FD001 and 210 edges over 21 sensors for FD002 and FD004, so the topology is not hand-specified. The graph network is best on every suite at 2% of the data, at 36.5 against 51.0 for the MLP and 76.2 for the GRU on FD001, 44.0 against 62.2 on FD002, and 43.8 against 68.0 on FD004, so the prior survives regime mixing rather than only helping in the single-condition case. The regime-conditional variant improves on the plain graph under mixing, at 41.8 against 44.0 on FD002 at 2%, and carries no benefit on FD001, where its conditioning vector is vacuous, which is the physics check we wanted. At full scale the flexible flat models overtake, at 21.7 against 19.4 on FD001 and 41.8 against 23.1 on FD002, and that reversal is the point rather than an embarrassment, because a structural prior should be worth most where data is scarce and is not a universal replacement for high-capacity sequence models.

Result 2: the same graph transfers, generalizes in size, and gives early warning. A message-passing network trained on a DC-motor machine family transfers zero-shot to an AC-motor family with identical interfaces, reaching next-step RMSE 3.16 against 7.23 for a flat MLP, a factor of 2.3, while in-distribution performance is comparable at 0.59 and 0.58. The identical trained model also applies unchanged to machines with 9 and 5 operators, at RMSE 0.56 and 0.47, where the flat MLP cannot be applied at all because its input dimension is fixed, while an edge-ablated twin given the

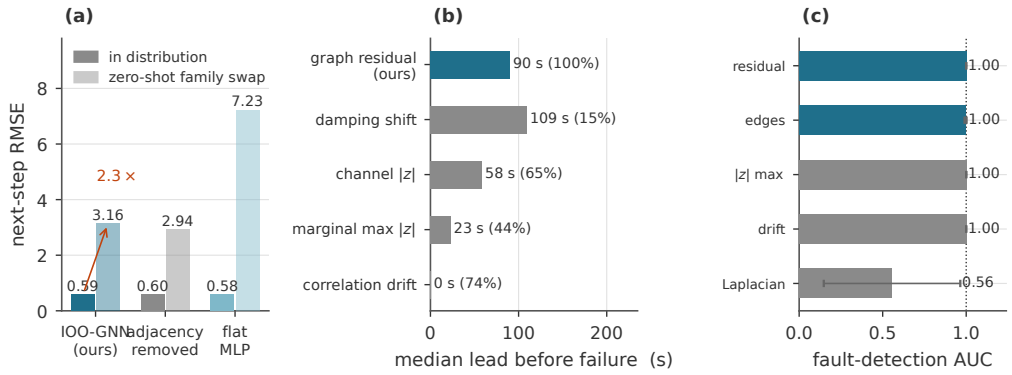


Figure 2: Transfer, early warning, and an honest tie. Panel (a) is next-step RMSE in distribution and under a zero-shot swap of the machine family, for a network trained on a DC-motor family; panel (b) is the median detection lead over 34 simulated runs with the wear channel hidden, with the detection rate in parentheses; panel (c) is real CWRU bearing vibration, 9 fault files against 3 healthy files measured by four health statistics.

same windows and no adjacency is indistinguishable from the graph network in distribution at 0.60 and realizes no topology-driven benefit.

The early-warning result, Figure 2b, is where the geometry does something no marginal statistic can. On simulated runs where a bearing-wear fault ramps in mid-run and the wear channel is hidden from every detector, a graph-structured conditional residual, which regresses each channel on its graph parents and thresholds at the healthy 99th percentile, detects *every* failure, 34 of 34, a median of 89.7 seconds before functional failure. Correlation drift, the standard geometric health metric, detects 74% of failures and fires essentially at failure with a median lead of 0 seconds. The residual responds to changes in coupling strength, and marginal and correlation statistics are blind to exactly those changes. On real hardware the method ports intact, since on Case Western Reserve University 12 kHz bearing vibration, 9 fault files against 3 healthy files, the same graph-health architecture of 16 band-energy operators with a coupling graph estimated on a healthy reference separates every fault file at AUC 1.0, as Figure 2c reports. We frame that as transferability evidence rather than superiority, because the marginal maximum-$|z|$ baseline also reaches 1.0 on these gross seeded faults [Smith and Randall, 2015]. Replacing partial correlation with the non-parametric conditional-mutual-information estimator gives a conservative graph of 11 edges against 105, and we report how it predicts in the audit below.

Auditing the effects. Three checks support these results. Removing the adjacency with the same windows erases the low-data advantage, since the no-edge control is 20.5 cycles worse than the graph network on FD002 at 2% and identical at saturation, so topology rather than capacity drives it. Pushing the conditional residual above its healthy threshold by a bounded gradient search needs about 0.23 z-units under an L_∞ budget, so the detector is not trivially corruptible. Training on the non-parametric graph gives 39.6 RMSE against 40.3 for the no-topology control and 36.5 for the Gaussian graph, so both graphs beat the control and the conservative one is consistent with the Gaussian one rather than contradicting it.

4 Discussion and limitations

The framework reframes the modeling of physical assets from flat data injection to geometric reading, where one graph encodes coupling structure and the same learned object drives cross-machine transfer, health monitoring, and eventually control. Its weakness is visible in our own results and worth stating plainly, which is that the structural prior is most valuable where data is scarce and is not a universal replacement for high-capacity sequence models once data is plentiful, so a hybrid of a graph term and a sequence residual on the de-coupled channels is the natural synthesis and is not built here. The graphs learned on real telemetry are statistical rather than hand-authorized physical catalogs, and the early-warning result rests on simulated fault ramps with the wear channel hidden by construction. Discovering operator catalogs from fleets of machines and registering interface signatures from design files would turn this from a discovered graph into a declared one, and that is the direction we would take next.

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Availability of code and data

The C-MAPSS and CWRU datasets are public, and the graph discovery, training, audit and figure code are available from the author on request together with the per-suite result files and the commands that regenerate every number and figure. The audit that bounds how much perturbation the detector tolerates is run as code rather than asserted in prose.

Use of AI tools

Generative AI tools were used for review and polish rather than for drafting: to audit the manuscript, to check every reported number against the committed per-suite result files and the reproduction pipeline, to run the independence-style audits as code rather than asserting them, and to verify citations against their sources. The authors directed the work and take responsibility for all content.